

School of Computing and Information Technology

Database Design CMP2018 (Practical)

Group Project – PART 2 (LAB GAMMA)

General Instructions

1. This lab is an individual assessment.
Although the ERD was completed in groups, each student must complete and submit their own SQL implementation of the assigned scenario.
2. You are expected to demonstrate your understanding of data integrity rules through the proper use of constraints.
3. Your SQL script must execute without errors in your SQL environment (SQLSERVER, MySQL, PostgreSQL, or Oracle).
4. Include comments (--) to label each section of your script.
5. Marks (6 + 5 + 5 + 7 + 6 + 5 + 6)

Scenario 1: Greg's Extreme Fitness Centre

- i) Using an INNER JOIN display all branches, their opening hours, facilities, trainers who work at that branch and the various payment plans that are offered.
- ii) Using a LEFT JOIN, display all female members who do not have a personal trainer.
- iii) Using a RIGHT JOIN, display all members who did not enter any competition.
- iv) Using Sub Queries and a Join statement, show all the members and trainer's name for those members who has a BMI in excess of 25 ($BMI = 703 \times \text{weight (lbs)} / (\text{height} * \text{height})$) and who have not been to the gym in the last 5 days.
- v) Using Sub-queries and an IN statement, display the trainer with the cheapest training fee, showing their name, contact details, and their BMI
- vi) Using a FULL Join, show all members, their payment plans, their trainers, their attendance record, any competitions entered and position placed in the competition.
- vii) Create a view that shows delinquent memberships. Display the members' details,

branch registered, type of plan, total amount paid, expiry date of the plan, balance

Prepared by: D. Allen Semester 1 AY 2025-26

outstanding, last payment date and next payment date for all members with a monthly plan.

Scenario 2: Elegant Events

- i) Using an INNER JOIN, display the event name, client name, event date, venue, event type, assigned staff (full name, gender, role), and the supervisor assigned to oversee the event.
- ii) Using a LEFT JOIN, display all staff members who have not been assigned to work on any event. Show full name (as one string), years employed, gender, specialty (event planning, décor, logistics, etc.), and phone number. Sort results by years employed, descending.
- iii) Using a RIGHT JOIN, display all venues that have not hosted any event for the last two (2) years.
- iv) Using **subqueries** and **JOIN statements**, display all events that were completed within budget. Show the client name, event type, budgeted amount, actual spending, event manager's name, years of experience, and event duration.
- v) Display the **first staff member ever hired** by Elegant Events. Show name, gender, years employed, date hired, age, and specialty. Your query **must** include at least one subquery and one **IN** statement.
- vi) Using a FULL JOIN, display all staff, their skills, the events they are assigned to, and the supervisor for those events.
- vii) Create a **VIEW** to show each staff member's name, gender, hourly rate, potential monthly earnings (based on a 45-hour week + overtime at double rate), age, specialty, and number of events worked in the last 12 months.

Scenario 3: Zoo Jam

- i) Using an INNER JOIN, display each animal, its species, the cage it is currently assigned to, and the employee who loaded the animal into that cage.

Show the animal name, gender, DOB, health status, species common name, scientific name, diet, cage ID, location, cage type and employee's full name and role.

Prepared by: D. Allen Semester 1 AY 2025-26

- ii) Using a **LEFT JOIN**, display all employees who have not been assigned to any cage (they are not currently responsible for any cage). Show full name (as one string), role, age, address, mobile number(s). Sort by age descending.
- iii) Using a **RIGHT JOIN**, display all cages that have never housed an animal (no cage history records). Show cage ID, location, cage type, size, height and depth.
- iv) Using subqueries and joins: For all animals whose **cage history shows that they completed a cage stay** (animal was assigned AND later removed), display animal name, species name, cage ID, duration spent in cage, employee who loaded the animal (full name, role, years employed). Only include animals whose time spent in a cage meets an expected number of days defined in your query.
- v) Using sub-queries and an IN statement, display the last employee employed by the zoo, showing their full name, age, date employed, months employed, address and gender.
- vi) Using a full join that shows all animals and their offspring, species they belong to, cage they are assigned to, type of cage, the employee responsible for that cage and the employee's role.
- vii) Create a view that displays each employee's full name, age, number of years of employment, number of years to retirement, gender, cost per hour, number of hours worked, potential salary (assuming a 40 hour work week, overtime is 2X the cost per hour)

Scenario 4: Kollections Museum

- i) Using an **INNER JOIN** display all, artwork currently assigned to exhibitions, including details about the exhibition and the curator solely responsible for managing the artwork. Show Art title/description, estimated value, creation date, acquisition date, condition status, artist name, exhibition title, theme, room, exhibition start and end dates, and curator's full name, gender, and date employed. Sort by exhibition title within exhibition start date.
- ii) Using a **LEFT JOIN**, display all art pieces that are **NOT** part of any exhibition showing description, epoch, culture of origin/country, estimated value.
- iii) Using a **RIGHT JOIN**, display all artworks on exhibitions that are NOT on loan from any other private entities or institutions.

- iv) Using **subqueries** and **JOINS**, display all artworks borrowed from lenders *and returned on time*. Show the art ID, description, lender name, lender type, date borrowed, returned date, duration (calculated), exhibition(s) where the art was displayed, and the curator.
 - v) Using a **subquery and an IN statement**, show the first employee to be employed at the museum. Show their name, date of employment, years of employment, address, and gender.
- Prepared by: D. Allen Semester 1 AY 2025-26
- vi) Using a **FULL JOIN**, display the combined list of all artwork, all exhibitions, and all employees assigned to exhibitions.
 - vii) Create a **VIEW** that shows all curators and their exhibitions. Display their full name, gender, date employed, years employed, DOB, Age, number of years left for retirement, number of exhibitions they have worked on, and the date of last exhibition.

Scenario 5: Football Federation

- i) Using an **INNER JOIN**, display information on the players who play in the Manning Cup Display their name, shirt number, position the team they belong to, the coach, matches they participated in, the number of minutes they played in the match, number of fouls, assist, red and yellow cards given in the match as well as the referee for the match. Sort the information by player name within teams.
- ii) Using a **LEFT JOIN**, show players who were **not selected** for any match during the 2024–2025 season (or you may select your own season criteria). Include age, position, nationality, and contract details.
- iii) Using a **RIGHT JOIN** display all the referees in the federation. Show their names and contact details. (Note, referees are players who do not belong to a team)
- iv) Using **subqueries** and **JOINS**, display all matches completed within regulation time (i.e. 90 minutes). Show the team, opponent team, score, head coach, match duration, referee, and assistant referees. (You may need to alter your table to include time the match started and ended if not included).
- v) Using a subquery and an IN statement, display the details of the youngest Premier League player in the league. Display their name, age, skill level, and shirt number.
- vi) Using a **FULL JOIN**, display all players, their positions, teams, coaches, injury records and matches they played in, fouls, assists, red and/or yellow cards received as well as the referees for those matches. Rename your column headings in the display to distinguish coaches, players, and referees.

- vii) Create a **VIEW** that shows players who were injured in a match. Display their names, age, shirt number, team belonged to, number of months with the team, injury records, match played, minutes played in a match and any cards given during the match.

Prepared by: D. Allen Semester 1 AY 2025-26

Scenario 6: Promaneng Company Limited

- i) Display the client, project location, date project began, duration, value, and employees assigned to the project with specific skills (you may choose), showing date assigned to the project, their gender, position, address, and the name of the employee who is overseeing the work on the project.
- ii) Using a left join, display the employees who have not been assigned to work on any project. Show their full name (as one string), age, skills, gender, and phone number. Sort the records by age.
- iii) Using a right join, display all equipment that is not currently being used on any projects.
- iv) Using sub-queries and join statements, for projects that were completed (project has ended) within the estimated time, we want to know the client, the value of the project, the employee who supervised the other employees that worked on the project (full name, position, years of employment), and the actual duration of the project.
- v) Show the employee who was the first to join the company. Show their name, gender, years of employment, date joined, dob, age, and their skills. Ensure that at least one sub-query and one IN statement are included in your solution.
- vi) Using a full join, display all employees, their skills, and the projects they are assigned to, as well as assigned work supervisors.
- vii) Create a view that shows each employee's name, rate per hour, gender, Potential earnings (based on a 48 hr work week with overtime paid at double the rate), date of birth, age, and number of years of employment for all employees assigned to work on projects. The potential earnings are based on their duration on a project to date. Assume a 30-day period for each month.

SUBMISSION REQUIREMENTS

Your submission must include:

1. The **SQL script file (.sql)** containing all implementation steps.

2. A **screenshot or exported output** showing that all statements executed successfully.
3. A brief **reflection paragraph** describing what you learned from any two queries that you thought were challenging.

All submissions must be uploaded by the deadline announced by your tutor/lab technologist. Late submissions will be subject to penalties as outlined in the course policy.

Prepared by: D. Allen Semester 1 AY 2025-26